

How to handle the ML Pole Problem: A poll

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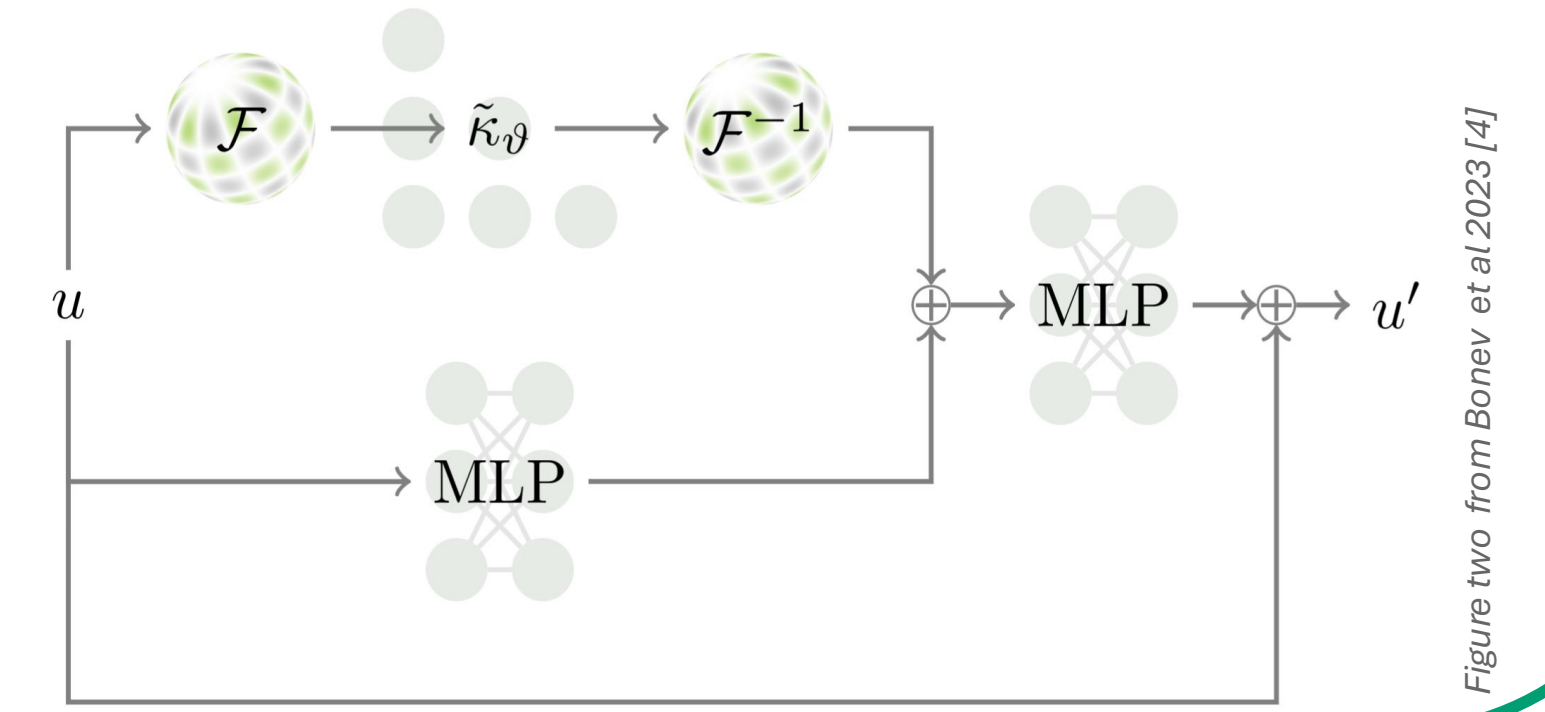
The ML Pole Problem

When working with climate data in the Arctic, we must consider how to handle the pole. In my case, the issue comes from using a CNN on latitude longitude grided data from 45°N to the pole. Here, the date line is a discontinuity, each grid cell has a different area, and at the pole kernel padding is hard. I here present the four solutions to deal with the pole.

Vote in the poll on how to handle the pole!

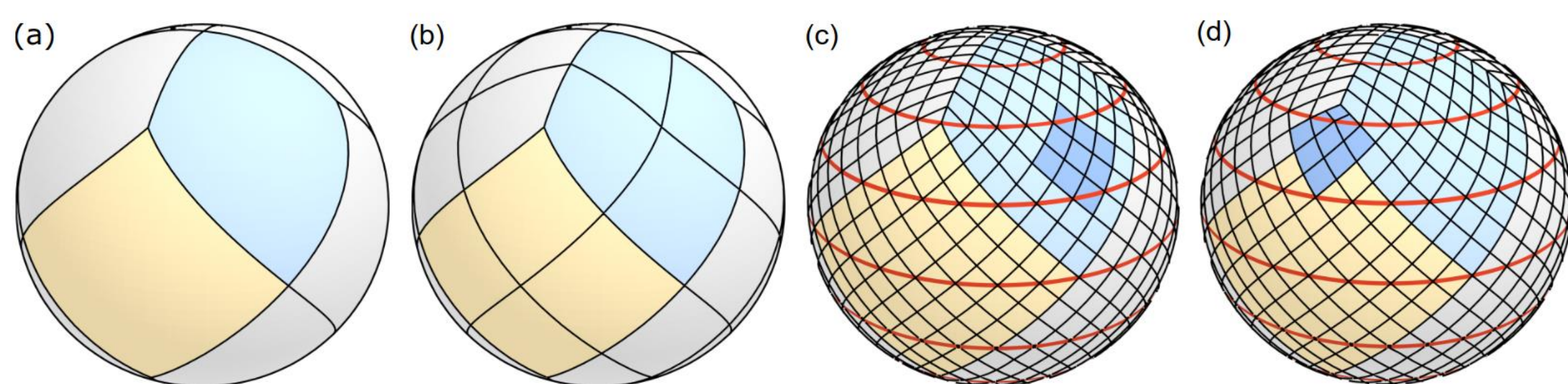
SF Neural Operators

The numerical pole problem was solved by using Spherical Fourier (SF) analysis. This can also be done with ML using SF neural operator blocks consisting of two MLPs and an SF part see figure. The input is split in two information streams one to an MLP and one to the SF part that applies a filter. An MLP then combines them.



HEALPix mesh

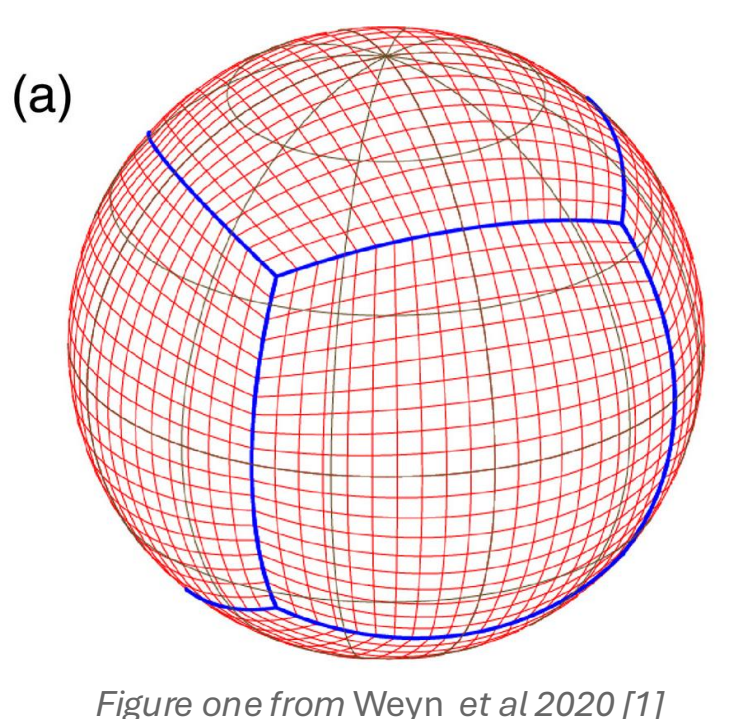
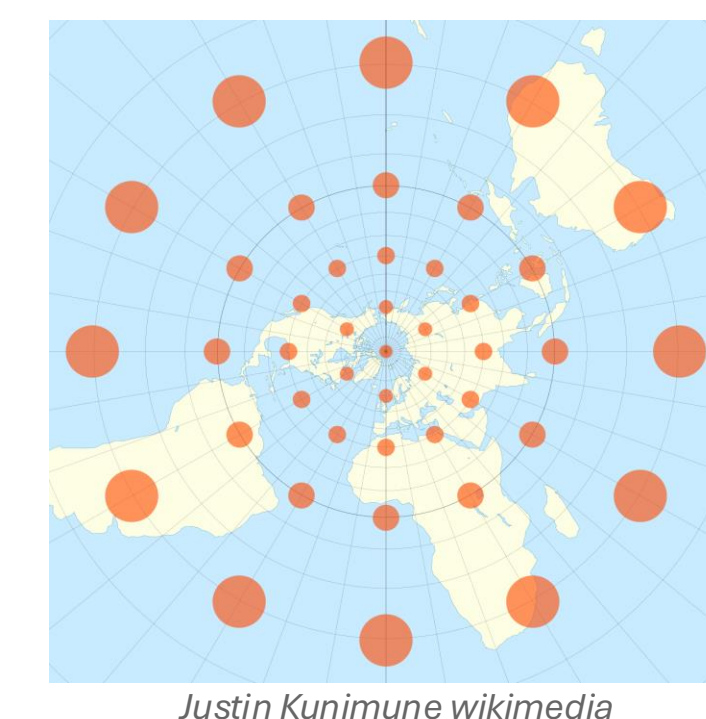
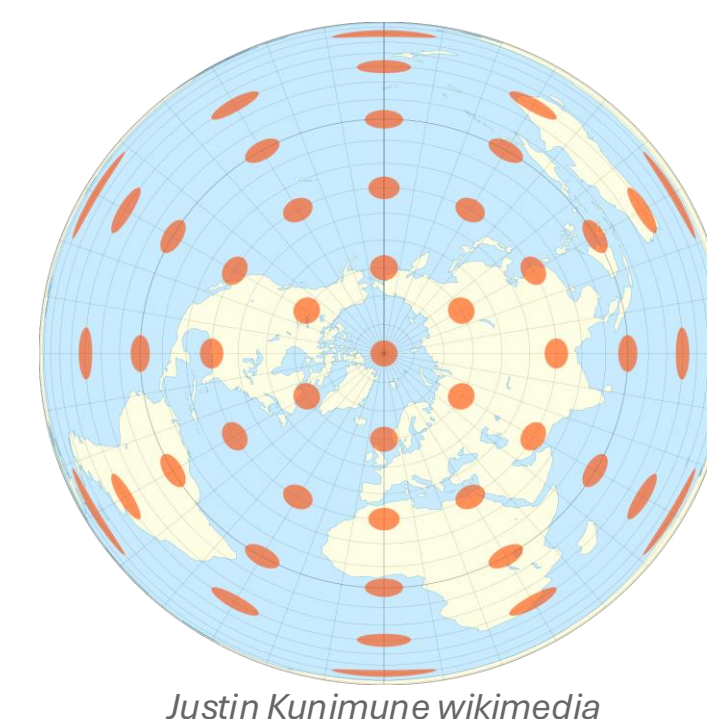
An equal area decomposition that allows for a classical CNN structure. The figure below shows the mesh. To construct the HEALPix mesh; first split the globe in polar-mid-latitude and tropical tiles then subdivide these. However, this requires interpolation at the corners.



Alternative Projections

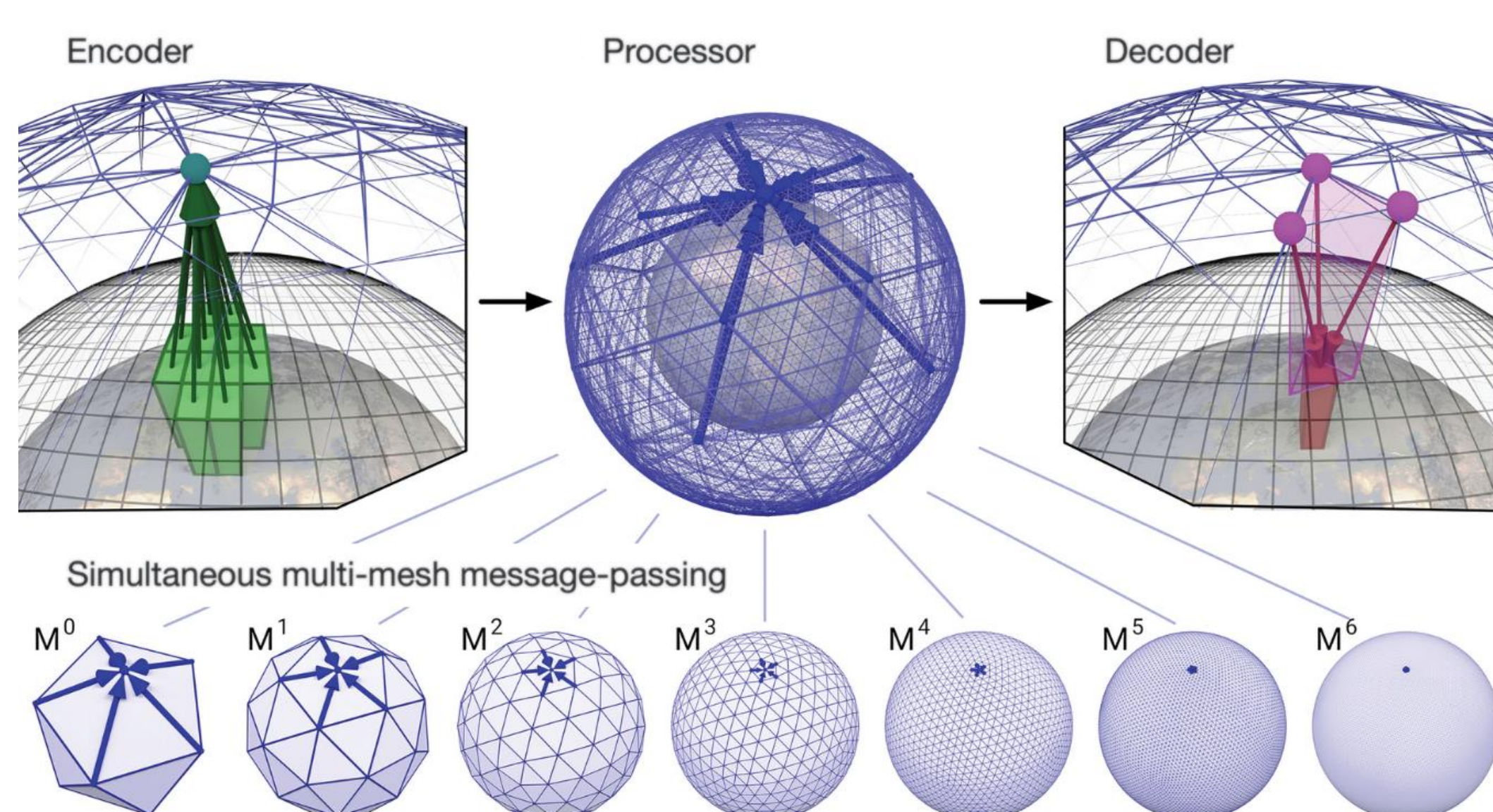
If we are only interested in the area between 45°N and the pole, a new projection could alleviate problem, and we could keep using the CNN. Here are three suggestions which should have minimal distortion and area change.

Lambert azimuthal Polar Stereographic Cubed Sphere



Hierarchical GNN

Inspired by GraphCast [5] and the ICON model. Here the data is on an equal area multi-scale icosahedral mesh that is passed to a graph neural network (GNN). The GNN integrates message passing with an encoder-processor-decoder architecture. This enables the model to learn atmospheric processes across different spatial extents. The icosahedron is almost equal-area and allows for specific targeting of the long-range process.



VOTE IN THE POLE POLL

Write your name in a box below

HEALPix mesh

SF Neural Operators

Hierarchical GNN

Alternative Projections

References:

- Weyn, J. A., Durran, D. R., & Caruana, R. (2020). Improving Data-Driven Global Weather Prediction Using Deep Convolutional Neural Networks on a Cubed Sphere. *J.o.A. in Modeling Earth Systems*, doi.org/10.1029/2020MS002109.
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Other Ideas?